

OPTIMIZATION OF ZF

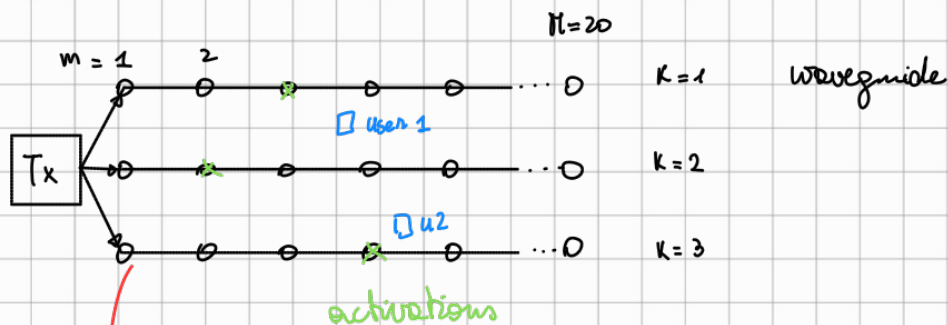
- sum-rate with ZF precoding ($K=3, U=2$)

$$R = \sum_k \log_2 \left(1 + \frac{E_k}{\sigma_n^2} \frac{|\underline{h}_k^H \underline{f}_k|^2}{\|\underline{f}_k\|^2} \right)$$

with $\underline{F} = \underline{H}(\underline{H}^H \underline{H})^{-1}$

$$= \sum_k \log_2 \left(1 + \frac{E_k}{\sigma_n^2} \cdot \frac{|\underline{h}_k^H \underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_k|^2}{\|\underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_k\|^2} \right)$$

- scenario



$\underline{S}$ (elements $S_{k,m}$) activation matrix s.t. $\begin{cases} S_{k,m} = \{0, 1\} \\ \sum_{m=1}^M S_{k,m} = 1 \rightarrow 1 \text{ activation per waveguide} \end{cases}$

- problem:

$$\underset{\underline{S}}{\text{maximize}} \{R\} = \underset{\underline{S}}{\text{minimize}} \{-R\}$$

s.t. C1: $S_{k,m} = \{0, 1\}$

C2: $\sum_m S_{k,m} = 1 \quad \forall k$

- relaxation of binary constraint C1 to continuous values

$\overline{C1}: 0 \leq S_{k,m} \leq 1$

$\Rightarrow \underset{\underline{S}}{\text{minimize}} \{-R\} \quad \text{s.t. } \overline{C1}, C2$

JOINT OPTIMIZATION FOR PA POSITION AND BEAMFORMING WEIGHTS

- given number of users U , number of waveguide K and M possible PA positions on every waveguide
- given the channel gain matrix $\underline{C}$ which contains the channel gain coefficients for every antenna position, for every waveguide, for every user:

$$\underline{C} \in \mathbb{C}^{K \times N \times U} \rightarrow \text{elements } C_{k,m,u}$$

$$\text{or } \underline{C} = [\underline{C}_1 \dots \underline{C}_U] \rightarrow \underline{C}_u \in \mathbb{C}^{K \times N}$$

- the channel matrix $\underline{H} \in \mathbb{C}^{K \times U}$ can be obtained through the binary selection matrix $\underline{S} \in \{0,1\}^{K \times N}$. Each channel vector $\underline{h}_u \in \mathbb{C}^{K \times 1}$ for each user is defined as

$$\underline{h}_u = \underline{S} \odot \underline{C}_u \cdot \underline{1}_N$$

and the channel matrix and its elements can be written as

$$\underline{H} = [\underline{h}_1 \dots \underline{h}_U] \quad H[k,u] = \underline{S}[k,:] \cdot \underline{C}[k,:,u]^T$$

- the signal model for user u , with its beamforming vector $\underline{w}_u \in \mathbb{C}^{K \times 1}$, is

$$\begin{aligned} y_u &= \underline{h}_u^H \underline{w}_u x_u + \underline{h}_u^H \sum_{i \neq u} \underline{w}_i x_i + n_u \\ &= (\underline{S} \odot \underline{C}_u \cdot \underline{1}_N) \cdot \sum_{i=1}^U \underline{w}_i x_i + n_u \end{aligned}$$

- SINR for user u and the sum rate are

$$\text{SINR: } \gamma_u = \frac{|\underline{h}_u^H \underline{w}_u|^2}{\sum_{i \neq u} |\underline{h}_u^H \underline{w}_i|^2 + \sigma_n^2} = \frac{|(\underline{S} \odot \underline{C}_u \cdot \underline{1}_N)^H \underline{w}_u|^2}{\sum_{i \neq u} |(\underline{S} \odot \underline{C}_i \cdot \underline{1}_N)^H \underline{w}_i|^2 + \sigma_n^2}$$

$$R = \sum_u \log_2(1 + \gamma_u)$$

- the optimisation problem is defined as:

$$\underset{\underline{S}, \underline{W}}{\text{maximization of } R} = \underset{\underline{S}, \underline{W}}{\text{minimization of } -R} \quad \text{s.t.}$$

$$C1: S_{k,m} \in \{0,1\} \xrightarrow{\text{relaxation}} \overline{C1}: 0 \leq S_{k,m} \leq 1$$

$$C2: \sum_m S_{k,m} = 1 \quad \forall k$$

$$C3: \sum_u \|\underline{w}_u\|^2 = \|\underline{W}\|_F^2 \leq P$$

the problem is solved with alternating optimization

→ fix $\underline{S}$ and optimise on $\underline{W}$

→ fix $\underline{W}$ and optimise on $\underline{S}$

→ until convergence for R

OPTIMIZATION OF $\underline{W}$

$$\underline{\Delta} = \underset{K \times 1}{\underline{W}} \underset{K \times U}{\underline{X}} = \sum_u \underset{U \times 1}{\underline{w}_u} x_u$$

$$\underset{U \times 1}{\underline{y}} = \underset{U \times K}{\underline{H}} \underset{K \times 1}{\underline{\Delta}} + \underset{U \times 1}{\underline{n}} = \underset{U \times U}{\underline{H}} \underset{U \times U}{\underline{W}} \underset{U \times 1}{\underline{x}} + \underset{U \times 1}{\underline{n}}$$

for user u

$$y_u = \underline{h}_u^H \underline{W} \underline{x} + n_u = \underline{h}_u^H \underline{w}_u x_u + \sum_{i \neq u} \underline{h}_u^H \underline{w}_i x_i + n_u = \underline{h}_u^H \underline{w}_u + \hat{n}_u$$

↳ effective noise: $\hat{n}_u = \sum_{i \neq u} \underline{h}_u^H \underline{w}_i x_i + n_u$

$$\begin{aligned} E\{\hat{n}_u\} &= E\left\{\sum_{i \neq u} |\underline{h}_u^H \underline{w}_i x_i|^2\right\} + E\{|n_u|^2\} \\ &= \sum_{i \neq u} |\underline{h}_u^H \underline{w}_i|^2 \underbrace{E\{|x_i|^2\}}_{=1} + \sigma_n^2 \\ &= \sum_{i \neq u} |\underline{h}_u^H \underline{w}_i|^2 + \sigma_n^2 = \sigma_{\hat{n}}^2 \end{aligned}$$

$$R_u = \log_2(1 + \gamma_u) = \log_2\left(1 + \frac{|\underline{h}_u^H \underline{w}_u|^2}{\sum_{i \neq u} |\underline{h}_u^H \underline{w}_i|^2 + \sigma_n^2}\right)$$

$$= \log_2\left(1 + \frac{|\underline{h}_u^H \underline{w}_u|^2}{\sigma_{\hat{n}}^2}\right) = \log_2\left(\frac{\sigma_{\hat{n}}^2 + |\underline{h}_u^H \underline{w}_u|^2}{\sigma_{\hat{n}}^2}\right)$$

- at the receiver: $\hat{x}_u = a_u y_u = a_u (\underline{h}_u^H \underline{w}_u x_u + \tilde{n}_u)$
- using a MMSE filter a_u^{MMSE} at the receiver for user u , s.t.

$$a_u^{\text{MMSE}} = \underset{f_u}{\operatorname{argmin}} \{ E \{ |\hat{x}_u - x_u|^2 \} \}$$

$$= \underset{f_u}{\operatorname{argmin}} \{ E \{ |a_u y_u - x_u|^2 \} \}$$

$$\begin{aligned} \rightarrow E \{ |a_u (\underline{h}_u^H \underline{w}_u x_u + \tilde{n}_u) - x_u|^2 \} \\ &= E \{ |(a_u \underline{h}_u^H \underline{w}_u - 1) x_u - a_u \tilde{n}_u|^2 \} \\ &= E \{ |(a_u \underline{h}_u^H \underline{w}_u - 1) x_u|^2 \} + E \{ |a_u \tilde{n}_u|^2 \} + 0 \quad \rightarrow \text{since } E \{ x_u \tilde{n}_u^* \} = 0 \\ &= E \{ |a_u \underline{h}_u^H \underline{w}_u - 1|^2 |x_u|^2 \} + a_u^2 \sigma_{\tilde{n}}^2 \\ &= |\underline{h}_u^H \underline{w}_u|^2 + 1 - 2 \operatorname{Re} \{ f_u \underline{h}_u^H \underline{w}_u \} + |a_u|^2 \sigma_{\tilde{n}}^2 \end{aligned}$$

$$\frac{d}{da_u} \stackrel{!}{=} 0 \rightarrow a_u |\underline{h}_u^H \underline{w}_u|^2 - (\underline{h}_u^H \underline{w}_u)^* + a_u \sigma_{\tilde{n}}^2 \stackrel{!}{=} 0$$

$$a_u (|\underline{h}_u^H \underline{w}_u|^2 + a_u \sigma_{\tilde{n}}^2) = (\underline{h}_u^H \underline{w}_u)^*$$

$$a_u^{\text{MMSE}} = \frac{(\underline{h}_u^H \underline{w}_u)^*}{|\underline{h}_u^H \underline{w}_u|^2 + \sigma_{\tilde{n}}^2} \quad \star$$

- the MSE for user u can be written as:

$$\begin{aligned} e_u &= E \{ |a_u^{\text{MMSE}} y_u - x_u|^2 \} \\ &= E \{ |a_u^{\text{MMSE}} (\underline{h}_u^H \underline{w}_u x_u + \tilde{n}_u) - x_u|^2 \} \\ &= E \{ |(a_u^{\text{MMSE}} \underline{h}_u^H \underline{w}_u - 1) x_u + a_u^{\text{MMSE}} \tilde{n}_u|^2 \} \\ &= |a_u^{\text{MMSE}} \underline{h}_u^H \underline{w}_u - 1|^2 + |a_u^{\text{MMSE}}|^2 \sigma_{\tilde{n}}^2 \\ &= |a_u^{\text{MMSE}}|^2 |\underline{h}_u^H \underline{w}_u|^2 + 1 - 2 \operatorname{Re} \{ f_u (\underline{h}_u^H \underline{w}_u)^* \} + |a_u^{\text{MMSE}}|^2 \sigma_{\tilde{n}}^2 \\ \star \quad &= \frac{|\underline{h}_u^H \underline{w}_u|^2 \cdot |\underline{h}_u^H \underline{w}_u|^2}{| |\underline{h}_u^H \underline{w}_u|^2 + \sigma_{\tilde{n}}^2 |^2} + 1 - 2 \frac{|\underline{h}_u^H \underline{w}_u|^2}{|\underline{h}_u^H \underline{w}_u|^2 + \sigma_{\tilde{n}}^2} + \frac{|\underline{h}_u^H \underline{w}_u|^2 \cdot \sigma_{\tilde{n}}^2}{| |\underline{h}_u^H \underline{w}_u|^2 + \sigma_{\tilde{n}}^2 |^2} \end{aligned}$$

$$\begin{aligned}
& \frac{|h_u^H \underline{w}_u|^4 + (|h_u^H \underline{w}_u|^2 + \sigma_n^2)^2 - 2|h_u^H \underline{w}_u|^2(|h_u^H \underline{w}_u|^2 + \sigma_n^2) + |h_u^H \underline{w}_u|^2 \sigma_n^2}{(|h_u^H \underline{w}_u|^2 + \sigma_n^2)^2} \\
& = \frac{\cancel{|h_u^H \underline{w}_u|^4} + \cancel{|h_u^H \underline{w}_u|^4} + \sigma_n^4 + 2\cancel{|h_u^H \underline{w}_u|^2 \sigma_n^2} - 2\cancel{|h_u^H \underline{w}_u|^4} - 2\cancel{|h_u^H \underline{w}_u|^2 \sigma_n^2} + |h_u^H \underline{w}_u|^2 \sigma_n^2}{(|h_u^H \underline{w}_u|^2 + \sigma_n^2)^2} \\
& = \frac{\sigma_n^2 (|h_u^H \underline{w}_u|^2 + \sigma_n^2)}{(|h_u^H \underline{w}_u|^2 + \sigma_n^2)^2} \\
& = \frac{\sigma_n^2}{|h_u^H \underline{w}_u|^2 + \sigma_n^2}
\end{aligned}$$

$$\hookrightarrow e_u^{-1} = 1 + \gamma_u$$

$$\Rightarrow R_u = \log_2 \left(\frac{\sigma_n^2 + |h_u^H \underline{w}_u|^2}{\sigma_n^2} \right) = \log_2(e_u^{-1}) = -\log_2(e_u)$$

Maximizing the sum rate is equivalent to minimize MSE using the MMSE receiver

$\hookrightarrow$ Problem 1: minimize $\left\{ \sum_n R_n \right\}$

Problem 2: minimize $\left\{ \sum_n \alpha_n e_n \right\}$ introducing MSE weights α_n

$\hookrightarrow$ comparing the two gradients w.r.t. $\underline{w}_u \rightarrow [\nabla_{\underline{w}_u}]_n = \frac{d}{d[\underline{w}_u]_n^*}$

$$\nabla_{\underline{w}_u} \left(\sum_n R_n - R_u(e_u) \right) = - \frac{\underline{h}_u \underline{h}_u^H \underline{w}_u e_u}{\sigma_n^2}$$

$$\hookrightarrow \nabla_{\underline{w}_u} R_u(e_u) = \nabla_{\underline{w}_u} \log_2(e_u) = \nabla_{\underline{w}_u} \log_2 \left(\frac{\sigma_n^2}{|h_u^H \underline{w}_u|^2 + \sigma_n^2} \right) =$$

$$= \underbrace{\nabla_{\underline{w}_u} \log_2(\sigma_n^2)}_{=0} - \nabla_{\underline{w}_u} \log_2(|h_u^H \underline{w}_u|^2 + \sigma_n^2) =$$

$$\hookrightarrow \nabla_{\underline{w}_u} |h_u^H \underline{w}_u|^2 = \nabla_{\underline{w}_u} \underline{w}_u^H \underline{h}_u \underline{h}_u^H \underline{w}_u = \underline{h}_u \underline{h}_u^H \underline{w}_u$$

$$= - \underline{h}_u \underline{h}_u^H \underline{w}_u \left(|h_u^H \underline{w}_u|^2 + \sigma_n^2 \right)^{-1} \cdot \frac{1}{\ln(2)} \cdot \frac{\sigma_n^2}{\sigma_n^2} = - \frac{\underline{h}_u \underline{h}_u^H \underline{w}_u e_u}{\sigma_n^2}$$

$$\hookrightarrow \nabla_{\underline{w}_u} R_i(e_i) = 0$$

$$\nabla_{\underline{w}_u} v_u e_u = \nabla_{\underline{w}_u} v_u \sigma_n^2 \cdot \left(|\underline{h}_u^H \underline{w}_u|^2 + \sigma_n^2 \right)^{-1} = -v_u \sigma_n^2 \cdot \underline{h}_u \underline{h}_u^H \underline{w}_u \cdot \left(|\underline{h}_u^H \underline{w}_u|^2 + \sigma_n^2 \right)^{-2} \frac{\sigma_n^2}{\sigma_n^2}$$

$$= -v_u \frac{\underline{h}_u \underline{h}_u^H \underline{w}_u}{\sigma_n^2} e_u^2$$

$$\rightarrow -\frac{\cancel{\underline{h}_u \underline{h}_u^H \underline{w}_u}}{\cancel{\sigma_n^2}} e_u = -v_u \frac{\cancel{\underline{h}_u \underline{h}_u^H \underline{w}_u}}{\cancel{\sigma_n^2}} e_u^2$$

$$e_u = v_u e_u^2$$

$$\boxed{v_u = e_u^{-1}}$$

$$\rightarrow e_u = (1 + \gamma_u)^{-1} \rightarrow R_u = \log_2(e_u^{-1})$$

$$v_u = (1 + \gamma_u) \rightarrow R_u = \log_2(v_u)$$

derive expression for $\underline{w}_u$ in closed form:

$$\underline{w}_u = \underset{\underline{w}_u}{\operatorname{argmin}} \left\{ \sum_i v_i e_i + \lambda \left(\sum_i \|\underline{w}_i\|^2 - P \right) \right\} = \underset{\underline{w}_u}{\operatorname{argmin}} \left\{ J(\underline{w}_u) \right\}$$

$$\begin{aligned} J(\underline{w}) &= \sum_i v_i \left[1 + |\underline{a}_i|^2 |\underline{h}_i^H \underline{w}_i|^2 + |\underline{a}_i|^2 \sum_{j \neq i} |\underline{h}_j^H \underline{w}_i|^2 + |\underline{a}_i|^2 n_i - 2 \operatorname{Re} \left\{ (\underline{a}_i \underline{h}_i^H \underline{w}_i)^* \right\} \right] + \lambda \left(\sum_i \|\underline{w}_i\|^2 - P \right) \\ &= \sum_i v_i |\underline{a}_i|^2 \sum_j |\underline{h}_j^H \underline{w}_i|^2 + v_i |\underline{a}_i|^2 n_i - 2 v_i \operatorname{Re} \left\{ (\underline{a}_i \underline{h}_i^H \underline{w}_i)^* \right\} + \lambda \left(\sum_i \|\underline{w}_i\|^2 - P \right) \\ &= \sum_i v_i |\underline{a}_i|^2 \left[|\underline{h}_i^H \underline{w}_u|^2 + \sum_{j \neq u} |\underline{h}_j^H \underline{w}_j|^2 \right] - 2 v_u \operatorname{Re} \left\{ (\underline{a}_u \underline{h}_u^H \underline{w}_u)^* \right\} + \lambda \|\underline{w}_u\|^2 + \text{not depending on } \underline{w}_u \end{aligned}$$

$$\frac{\partial J}{\partial \underline{w}_u^*} \stackrel{!}{=} 0 \rightarrow \sum_i v_i |\underline{a}_i|^2 \underline{h}_i \underline{h}_i^H \underline{w}_u - v_u \underline{a}_u^* \underline{h}_u + \underline{w}_u \stackrel{!}{=} 0$$

$$\boxed{\underline{w}_u = v_u \underline{a}_u^* \left(\sum_i v_i |\underline{a}_i|^2 \underline{h}_i \underline{h}_i^H + \lambda \underline{I}_K \right)^{-1} \underline{h}_u}$$

alternate optimisation algorithm to maximize SR

initialize $\underline{W}^0$

update 1) $\underline{a}_u^n \mid \underline{W}^{n-1}$

2) $v_u^n \mid \underline{W}^{n-1}$

3) $\underline{W}^n \mid v_u^n, \underline{a}_u^n$

compute $R(\underline{W})$

$$\hookrightarrow a_u = \frac{(h_u^H w_u)^*}{|h_u^H w_u|^2 + \sigma_n^2} = \frac{w_u^H h_u}{\sum_i |h_u^H w_i|^2 + \sigma_n^2}$$

$$\hookrightarrow \underline{Q} = \text{diag}\{w^H H\} \cdot \left(\sum_u |H^H w|^2 + \sigma_n^2 \right)^{-1}$$

$$\hookrightarrow e_u = \frac{\sigma_n^2}{|h_u^H w_u|^2 + \sigma_n^2} = \frac{\sum_{i \neq u} |h_u^H w_i|^2}{\sum_i |h_u^H w_i|^2 + \sigma_n^2} = \frac{\sum_i |h_i^H w_i|^2 - |h_u^H w_u|^2}{\sum_i |h_u^H w_i|^2 + \sigma_n^2} = 1 - \frac{|h_u^H w_u|^2}{\sum_i |h_u^H w_i|^2 + \sigma_n^2}$$

$$\hookrightarrow \underline{e} = \underline{1}_U - \frac{|\text{diag}\{H^H w\}|^2}{\sum_i |h_u^H w_i|^2 + \sigma_n^2} = \underline{1}_U - |\text{diag}\{H^H w\}|^2 \cdot \left(\text{Tr}\{|H^H w|^2\} + \sigma_n^2 \right)^{-1}$$

$$\hookrightarrow \underline{w}_u = v_u a_u^* \left(\sum_i v_i |a_i|^2 h_i h_i^H + \lambda I_K \right)^{-1} h_u$$